Q.1.

i. If R is the set of all integers and if $a * b = a + b + 3$. Show that $(R, *)$ is an abelian group.

Ans/ To show that $(R, *)$ group, it must satisfy five properties:

① Binary Property:

For any $a, b \in R$ since a and b are integers, then $a+b+3$ is also an integer.

Therefore $a * b \in R$ holds true.

② Associative Property:

For any $a, b, c \in R$:

$(a * b) * c = (a+b+3) * c = a+b+3+c+3 = a+b+c+6$

$a * (b * c) = a * (b+c+3) = a+b+c+3+3 = a+b+c+6$

$\therefore (a * b) * c = a * (b * c)$

$\therefore$ the operation is associative

③ Identity Property / Element:

Let $e \in R$ be the identity element such that:

$$a * e = a$$

$$a + e + 3 = a \Rightarrow e = -3$$

$\therefore -3 \in R$, the identity element exists and is $e = -3$

④ Inverse Element:

Let $a^{-1} = a'$ be the inverse of a such that:

$$a * a' = e = -3$$

$$a + a' + 3 = -3 \Rightarrow a' = -6 - a$$

$\because a$ is an integer, $-6 - a \in R$, Thus, every element has an inverse.

⑤ Commutative Property:

For any $a, b \in R$:

$$a * b = a + b + 3 = b + a + 3 = b * a$$

$\therefore$ The operation is commutative

Since all five properties holds true, thus $(R, *)$ is an ~~albe~~ abelian group.

Q.1.
ii. Show that $G = \{1, -1, i, -i\}$ is a cyclic group under multiplication

Ans) A group G is cyclic if there exists at least one element $g \in G$ such that every element of G can be expressed as an integer power of g.

Powers of i:

$i^1 = i$

$i^2 = -1$

$i^3 = i^2 \cdot i = -i$

$i^4 = 1$.

The set generated by i is $\{i, -1, -i, 1\} = G$.

Similarly for $-i$:

$(-i)^1 = -i$

$(-i)^2 = -1$

$(-i)^3 = i$

$(-i)^4 = 1$

The set generated by $-i$ is $\{-i, -1, i, 1\} = G$

Since G can be generated by a single element, $G = \{1, -1, i, -i\}$ is a cyclic group under multiplication.

Q.1.
iii. Show that $(Z_4, +)$ and $(Z_5^*, .)$ are isomorphic.

Ans) $(Z_4, +_4)$ has elements $\{0, 1, 2, 3\}$ under addition modulo 4.

It is cyclic group of order 4 generated by 1:

$1^1 = 1, \quad 1+1 = 2, \quad 1+1+1 = 3, \quad 1+1+1+1 = 0$

$(Z_5^*, \cdot_5)$ has elements $\{1, 2, 3, 4\}$ under multiplication modulo 5

It is also a cyclic group of order 4 generated by 2:

$2^1 = 2 \pmod 5$ | $2^3 = 8 \equiv 3 \pmod 5$

$2^2 = 4 \pmod 5$ | $2^4 = 16 \equiv 1 \pmod 5$

Two finite cyclic groups of the same order are always isomorphic. we can define a bijective mapping $f : Z_4 \to Z_5^*$ by sending generator to generator ($f(k) = 2^k \pmod 5$):

- $f(0) = 2^0 \pmod 5 = 1$
- $f(1) = 2^1 \pmod 5 = 2$
- $f(2) = 2^2 \pmod 5 = 4$
- $f(3) = 2^3 \pmod 5 = 3$.

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- Bijective: Each element in Z_4 maps to a unique element in Z_5^*, and the mapping is both one-to-one and onto.
- Homomorphism: For any $x, y \in Z_4$:

$$f(x +_4 y) = 2^{x+y} \pmod 4 = 2^x \cdot 2^y \pmod 5 = f(x) \cdot_5 f(y)$$

Therefore, f is an isomorphism, thus $(Z_4, +) \cong (Z_5^*, \cdot)$

Q.2.

i. Consider (3,6) encoding function defined below $e: B^3 \to B^6$ defined by
$e(000) = 000000$; $e(001) = 000110$; $e(010) = 010010$; $e(011) = 010100$;
$e(100) = 100101$; $e(101) = 100011$; $e(110) = 110111$; $e(111) = 110001$.
Show that the encoding function is a group code.

Ans An encoding function $e: B^m \to B^n$ is a group code if and only if the set of codewords (the range/image of e, denoted by $C = e(B^m)$) forms a subgroup of $(B^n, \oplus)$.

Since C is a finite subset, it is a subgroup if and only if:

1. The identity element (000000) belongs to C (which it does, since e(000) = 000000).
2. C is closed under bitwise XOR addition ($\oplus$).

Alternatively, an encoding function is a group code if it is a homomorphism, i.e., $e(x \oplus y) = e(x) \oplus e(y)$ for all $x, y \in B^3$.

Let us express the codewords in terms of generator matrix formulation:

Notice that each codewords $e(b_1 b_2 b_3) = (c_1 c_2 c_3 c_4 c_5 c_6)$ has:

$c_1 = b_1$	$c_4 = b_1 \oplus b_3$
$c_2 = b_2$	$c_5 = b_2 \oplus b_3$
$c_3 = b_3$	$c_6 = b_1$

Let's check the parity bits for consistency:

Checking the parity bits for consistency

- For 001: $c_4 = 0 \oplus 1$, $c_5 = 0 \oplus 1 = 1$, $c_6 = 0 \Rightarrow 000110$ (matches)
- For 010: $c_4 = 0 \oplus 0 = 0$, $c_5 = 1 \oplus 0 = 1$, $c_6 = 0 \Rightarrow 010010$ (matches)

• For 100 : $c_4 = 1 \oplus 0 = 1$, $c_5 = 0 \oplus 0 = 0$, $c_6 = 1 \Rightarrow 100101$ (matches)

• For 111 : $c_4 = 1 \oplus 1 = 0$, $c_5 = 1 \oplus 1 = 0$, $c_6 = 1 \Rightarrow 111001$ (matches)

Since each output bit is a linear combination (mod 2) of the message bits, the map is linear:

$$e(x \oplus y) = e(x) \oplus e(y) \quad \forall\, x, y \in B^3$$

∴ the set of codewords $C = \text{Ran}(e)$ is closed under $\oplus$, making it a subgroup of B^6. Hence, the encoding function is a group code.

Q.2.
ii. Consider (3, 6) encoding function $e: B^2 \to B^6$ defined by $e(00) = 000000$; $e(10) = 101010$; $e(01) = 011110$; $e(11) = 111000$. How many errors can it detect.

Ans) The error detecting capability of a code depends on the minimum Hamming distance d_{min} between any pair of distinct codewords.

① Calculate the Hamming distances between all distinct pairs of codewords

$d(e(00), e(10)) = wt(101010) = 3$

$d(e(00), e(01)) = wt(011110) = 4$

$d(e(00), e(11)) = wt(111000) = 3$

$d(e(10), e(01)) = wt(101010 \oplus 011110) = wt(110100) = 3$

$d(e(10), e(11)) = wt(101010 \oplus 111000) = wt(010010) = 2$

$d(e(01), e(11)) = wt(011110 \oplus 111000) = wt(100110) = 3$

② Determine d_{min}:

$$d_{min} = \min(3, 4, 3, 3, 2, 3) = 2$$

③ Calculate error-detecting capability:

A code can detect up to k errors if and only if:

$$d_{min} \geq k + 1 \Rightarrow k \leq d_{min} - 1$$

$$k \leq 2 - 1 = 1$$

∴ the code can detect 1 error.

Q.2.
iii. $e(000) = 000000$; $e(001) = 000110$; $e(010) = 010010$; $e(011) = 010100$;
$e(100) = 100101$; $e(101) = 100011$; $e(110) = 110111$; $e(111) = 110001$.
Find minimum distance.

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Since this code is a group code (linear code), the minimum distance d_{min} is equal to the minimum Hamming weight of all non-zero codewords:

$$d_{min} = \min_{x \neq 000} wt(e(x))$$

Compute the weight (number of 1s) of each non-zero codeword:

- $wt(e(001)) = wt(000110) = 2$
- $wt(e(10)) = wt(010010) = 2$
- $wt(e(011)) = wt(010100) = 2$
- $wt(e(100)) = wt(100101) = 3$
- $wt(e(101)) = wt(100011) = 3$
- $wt(e(110)) = wt(110111) = 5$
- $wt(e(111)) = wt(110001) = 3$

The minimum weight among all non-zero codewords is:

$$d_{min} = \min(2, 2, 2, 3, 3, 5, 3) = 2$$

Therefore, the minimum distance is 2.